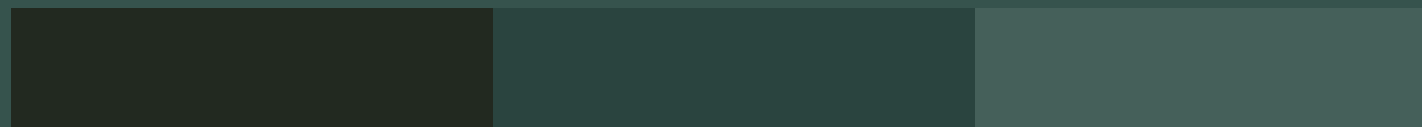


APPLICATION OF ML IN BIOTECHNOLOGY



Project Information

01

- The objective of this Summer Project is to introduce them to three main bioinformatics research areas through three projects – **Drug Discovery**, **DNA Sequencing** and **Diabetes Prediction** (Depending on the learning rate we will decide whether to do all 3 or not).
- Our aim is to provide the mentees with a comprehensive understanding of the core basics of ML, application of ML in the field of bioinformatics, including introduction to relevant research papers and biological databases.
- This project of ML and bioinformatics not only expands their horizons but also demonstrates the versatility of machine learning in addressing complex biological problems.
- **Target Audience** : Students who want to explore ML and it's application in bioinformatics. Well suited for learning ML from the core basics.



Relevance/Motivation:

01

- We all know the field of biotech is expanding all over the years and it is highly important for students to explore this field which caters the benefits from both biology and programming.
- Through this project students can explore how computational methods are revolutionizing fields like bioinformatics and contributing to groundbreaking discoveries in healthcare, genetics, and beyond. It will also include case studies through which they can explore the above described contributions.
- The main benefit is that this project focuses very much on the core basic and in-depth knowledge on ML which many students lack even after spending much time on learning. It will be most beneficial to students who want to explore this field and are facing difficulty in the starting phase of the learning.



Problem Statement:

- The project aims to introduce students to the field of bioinformatics through three major ML projects – Drug Discovery, DNA Sequencing and Diabetes prediction. It aims to solve the challenge of imparting foundational knowledge of ML and data analytics and most importantly its practical application in the field of bioinformatics.
- This project will help students in understanding key concepts involved in ML like how model works, EDA, model validation, fitting, algorithms of some models like random forest, decision trees, classification and regression.
- Data preprocessing : Techniques to handle missing values, categorical variables, cross validation, etc.
- Feature engineering : Mutual information metric, Creating features, K-means clustering, PCA, Target encoding, etc.



Applications

02

01

Drug
Discovery

02

Dna
Sequencing

03

Diabetes
Prediction

- Students will apply the knowledge gained to analyse the biological data and build models useful in some interesting challenges of bioinformatics.

Approach

02

We will approach in following steps:

- 1.Introduction to the field of **bioinformatics and ML**/Overview of the project
- 2.Fundamental Introduction of Google Collab/Jupyter Notebook, Python, Github, etc.
- 3.Modules on machine learning and libraries(Pandas, Numpy, Matplotlib, Seaborn)
- 4.Case studies, Articles and Research paper on relevant applications
- 5.Discussion on the topics coming ahead in the project
- 6.Applying the knowledge to build the model for :

DNA sequencing

Drug Discovery

Diabetes Prediction

- Relevant research papers and code files will be provided for the reference.
- We will provide necessary resources, including access to a programming environment with essential libraries. Tasks will be assigned periodically based on the requirement.
- Assignments need to be submitted at the end of every week by each student.
- Articles will be provided once a week related to widespread application of ML in biology



Logistics

03

1. Duration : **7-8 Weeks**
2. Work Load : **8-10 Hours**
3. No. of session per week : **3**
4. Preferred Batch for mentees : **Y23s**
5. Prerequisites : Enthusiasm to learn and complete the project!!
No prerequisites

If more applications : **Short case study(relevant to our project topic)** based assignment included in the application form of the project to make sure applicants make the least effort/1st step towards the project with interest. We will conduct a **test using a Google Form** containing 15-20 questions.

6. Expected Number of mentees – **(15-17)**
7. Mode of conduction and Communication channel : **Online and Whatsapp**



Tech Stacks and Evaluation

04

Programming language : Python

Libraries and Framework : Pandas, Numpy, Scikit-learn, Tensorflow

Tools : Github, Google Colab or Jupyter notebook

Attendance : If we plan to take 'n' sessions in the project, then a mentee must attend 'n-3' sessions to get evaluated further. Every student will need to submit code files (assignments) given to them at the end of every week in a common place like github or google drive.

Mid-Term Evaluation : Every student needs to make a report/presentation of the work they have done, describe their learnings till the mid of the project and what they will do ahead in the Presentation. This will be followed by a short viva session with the student. A short task will also be given to students

There will be a final project which needs to be completed to get ratified for the project.

End-Term Evaluation : Same as mid-term eval. Final report/presentation along with the submission of the final project. Github repository making at the end of the project which will include all the code files and a short description of the project.



Project Deliverables

- A good in-depth knowledge of data analysis, data cleaning and feature engineering analyses.
- Students will be well built with the basics of ML which includes model building, validation and data preprocessing.
- Good knowledge of various biological databases and along with this research paper reading habit which might be important for many students further ahead.
- Models for major applications like :
 - 1) DNA sequencing
 - 2) Drug Discovery
 - 3) Diabetes prediction



Timeline

WEEK 1 INTRODUCTION TO THE PROJECT

“How to setup google collab/Jupyter Notebook and kaggle”

Resources for Python, Git, Github and Introduction to ML will be provided

Students will have to upload files to a github repo using git

Assignment 1 : A python Assignment to make them comfortable with the language

Assignment 2 : ML based short assignment

Article/Research Paper on application of ML in bioinformatics field

WEEK 2 INTERMEDIATE MACHINE LEARNING

This will cover Missing values, categorical variables, Pipelines, Cross Validation and Introduction of algorithm and math of some models like Decision trees, Random forest, etc.

Assignment 3 : ML based short assignment

Assignment 4 : It may be about creating a report around the fundamental analysis of a Case study regarding ML application in biology and brainstorming

WEEK 3 FEATURE ENGINEERING

Feature creation, Some feature scoring metrics like Mutual information and K-best features, Some techniques like Target encoding, K-means Cluster, PCA, etc.

Assignment 5 : ML based short assignment

WEEK 4 PYTHON LIBRARIES AND ML MODEL

Matplotlib and Seaborn learning

Pandas and Numpy learning

Linear and Logistic regression

Assignment 6 : Exercises based on libraries

MID TERM EVALUATION

Updates in the form of **report/ppt** : This will include the things they have learnt so far and how will they apply it in the upcoming half of the project. It will also include a proper feedback from them regarding the project and difficulties faced by them. This will be followed by a short **QnA session with the mentee**.

Diabetes Prediction Task (information will be provided)

WEEK 5 Introduction to DNA sequences and Biological databases

Uniprot, ChEMBL, NCBI and reviewing **Research papers** regarding DNA sequencing

DNA sequencing ML based Model to predict the species of a particular sequence given

Dataset will be provided and brief introduction about “DNA Sequencing”

Data analysis and applying knowledge of intermediate ML and feature engineering

Model building using a basic classifier model

Assignment 7 : Code file submission

WEEK 6 Fine tuning and Model Validation

Experimenting the dataset predictions through various other model to increase accuracy

Starting the **Drug Discovery Project**, briefing about “Drug discovery”

Reviewing **Research paper** regarding Drug discovery
Dataset will be provided

Data analysis and applying knowledge of intermediate ML and feature engineering

WEEK 7 Final Project Completion

Completion of Drug Discovery model

Submission of all the code files and Report/ppt for Endterm eval

Assignment 7: Code file submission and Report/ppt

WEEK 8 Introduction to DL and Neural Networks(Optional/If time permits)

Introduction to **DL** and **Neural Networks**

Common Loss functions in ML

Backpropagation and **CNN**

END TERM EVALUATION

Updates in the form of **report/ppt** : This will include the things they have learnt so far and how will they apply it in the upcoming half of the project. It will also include a proper feedback from them regarding the project and difficulties faced by them. This will be followed by a short **QnA session with the mentee**.

[Resources for the project](#)

Ensuring Low-Mentee Dropout

07

- **Weekly assignments solving session :** By ensuring that there is a **assignment solving session** along with the normal sessions then the most doubts will get solved in that session itself and also they will not feel that interaction is less. Each student is required to submit their own assignment prior to the session, ensuring active participation and effort.
- **Workload less per week :** One main reason of dropping out is due to very much workload in projects and also some students whose grasping is low they feel overwhelmed by new things. To ensure fairness in this we can give them enough time for basics also because at least output from the project should be very strong. They should be capable of doing things even if they take time.
- **One Session for only feedback and normal talks:** This session can be arranged to take continuous feedback from the mentee regarding the project and also this will be a normal talks session to make sure connection is made throughout. Also, taking feedbacks after every session is a good way to tackle low mentee dropout





THANK YOU!

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